JEAN CLAUDE NIYOMUGABO

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PROFESSIONAL PROFILE

Agricultural systems researcher and builder focused on how farmers decide whether to trust and adopt digital and AI-enabled tools. I combine field experience with farmer cooperatives in Africa, research with producers across the U.S. Midwest, and hands-on work with irrigation sensors, data analysis (R, Stata, Python), and AI prototypes. Seeking industry roles in digital agriculture, agricultural AI research, and farmer-facing product and advisory work; also open to PhD programs.

EDUCATION

University of Nebraska–Lincoln, Lincoln, NE *Expected May 2027*

M.S., Biological Systems Engineering, Agricultural Systems Technology (Thesis Option) | CGPA 3.826/4.00

Thesis: Perceptions, Trust, and AI Readiness for AI-Enabled Agricultural Tools: A Cross-Sectional Study of Agricultural Producers and Students in the U.S. Midwest

Advisor: Dr. Santosh K. Pitla | *Committee:* Dr. Deepak Keshwani, Dr. Heather Akin, Dr. Guillermo R. Balboa

Relevant coursework: Statistical Methods for Research (STAT 801A; R and SAS); Research Methods in Leadership Education (ALEC 845); Agentic AI for Workflow Automation (AGST 492); Resilience Design in Agriculture (NRES/AGEN/BSEN 945); Irrigation Laboratory and Field Course (AGEN/AGST 854)

Rwanda Institute for Conservation Agriculture (RICA), Bugesera, Rwanda *August 2023*

B.Sc., Conservation Agriculture, specialization in Irrigation and Mechanized Agriculture | CGPA 3.37/4.00, Second-Class Honors (Upper Division); Howard G. Buffett Foundation Scholarship

Capstone: Evaluating Strawberry Yield in Vertical and Horizontal Hydroponic Farming Systems (Eza Neza)

Relevant coursework: Introduction to Agricultural Economics and Business; Intermediate Agricultural Business; Agricultural Business Planning; Enterprises and Marketing; Practical Mathematics for Agriculture; Applied Agriculture Research Methods; Agriculture and Environmental Physics; Site Specific Crop Management

RESEARCH EXPERIENCE

Researcher, Agricultural Water Management and Sensor-Based Irrigation, University of Nebraska–Lincoln 2026 – Present

- Conduct field research on 15 small urban farms (10 in Lincoln, 5 in Omaha) evaluating irrigation practices, water-use efficiency, crop productivity, and farm profitability under resource constraints comparable to those facing smallholder systems.
- Contribute to installation and monitoring of 60 soil-moisture sensors recording about every 15 minutes through the growing season; support instrumentation, data-quality checks, and farmer engagement.
- Link high-frequency field data to irrigation timing, farmer decision-making, and adoption of accessible, climate-adaptive water-management technologies.

Graduate Research Assistant, Cultivate ACCESS to Data Science in Agriculture (USDA-NIFA), University of Nebraska–Lincoln Aug 2025 – Present

- Lead the design of an IRB-approved, cross-sectional research study grounded in the Technology Acceptance Model and Diffusion of Innovations, measuring trust, perceived usefulness, compatibility, data concerns, training needs, and willingness to pay among farmers and next-generation producers. Data collection is ongoing.
- Designed a multi-state recruitment strategy targeting up to 600 farmers and agriculture students across 11 Midwestern states, with protocols to reduce fraudulent responses and protect participant privacy.
- Clean, manage, and analyze quantitative and qualitative data in R, Stata, Minitab, and Python, computing descriptive and inferential statistics to assess drivers of technology adoption.
- Contribute to extension materials for farmers, students, and extension professionals; mentor high school students in a USDA-funded pathway program connecting data science to agriculture.

Founder and Research Developer, AgiriteChat 2026 – Present

- Built an agentic AI agricultural assistant with retrieval-based knowledge for Nebraska and Rwanda's Eastern Province, voice input, and live weather data (LangGraph, Streamlit, Llama 3.3 70B). The prototype has not yet

been tested with farmers; I use it as a research testbed for localization, transparency, and farmer trust.

Student Advisor on Responsible AI Use, PRAIRIE Initiative, UNL College of Engineering Jan 2026 – Present

- Co-authored a multilevel framework for responsible AI integration in engineering education, informed by qualitative sentiment analysis of students, faculty, and industry partners.

FIELD AND PROFESSIONAL EXPERIENCE

Cooperative and Business Development Advisor, Agriterra, Kigali, Rwanda Jan 2024 – Jul 2025

- Advised 40 maize, soybean, rice, and potato cooperatives on technology adoption, mechanization, irrigation, water management, productivity, and youth participation across rural Rwanda.
- Coordinated demonstrations of tractors, center-pivot, drip, and flood irrigation, rainwater harvesting, and wastewater reuse; connected 200 farmers with mechanization providers.
- Coordinated training for 15 cooperative leaders in communication and business development, strengthening partnerships with the Netherlands Ministry of Foreign Affairs and private-sector donors.

Urban Agriculture Assistant, FAO Rwanda, Kigali, Rwanda Jan – Dec 2023

- Supported applied work at a UN FAO demonstration farm on high-value crops, hydroponics, automated drip irrigation, and wastewater reuse; contributed to data collection on smallholder beekeeping practices.
- Delivered four youth trainings on water management, wastewater reuse, hydroponics, and efficient irrigation; trainees later secured a UNDP Rwanda grant for a hydroponic greenhouse.

Co-Founder, Agirite, Kigali, Rwanda Aug 2021 – Jul 2025

- Led a social enterprise translating research and field experience into practical knowledge for farmers and young people. Trained over 3,000 young farmers on mechanized and sustainable agriculture with Hello Tractor; led the Buy from Youth market-linkage campaigns (2022–2024), connecting 200+ young farmers with urban markets.

Urban Farming and Sustainability Intern, Eza Neza, Rwanda Jan – Jul 2023

- Managed automated irrigation across four hydroponic greenhouses, optimizing water use at Rwanda's largest strawberry greenhouse operation; completed a capstone evaluating vertical and horizontal hydroponic yield.

PUBLICATIONS AND PRESENTATIONS

Stone, A. B., Stone, M. C., Bevins, A., Niyomugabo, J. C., Magara, I., Abaare, J., Heeren, D. M., & Abu Zouriq, M. (2026). Shared language for responsible AI integration: A framework for responsible AI use in engineering education. White paper, University of Nebraska–Lincoln Digital Commons.

Niyomugabo, J. C. (2026, July). Perceptions, trust, and AI readiness influencing adoption of AI-enabled agricultural tools among early adopters and next-generation producers in the U.S. Midwest. American Society of Agricultural and Biological Engineers (ASABE) Annual International Meeting.

Invited panelist, University of Nebraska System Graduate College Workshop, NU AI Institute (April 2026); invited panelist, PRAIRIE Initiative launch event, UNL College of Engineering (February 2026).

TEACHING AND MENTORING

Teaching Assistant, AGST/BSEN 416/816 Sensors and Control Systems, University of Nebraska–Lincoln Fall 2026

Cultivate Access Ambassador, RICA and University of Nebraska–Lincoln Jan 2022 – Jan 2023

- Mentored 50 Rwandan high school students pursuing agriculture in university.

Undergraduate Teaching Assistant, Biology, RICA Sept – Dec 2021

- Assisted 84 students with coursework, laboratory sessions, grading, and tutoring.

SKILLS

Statistics and data: R, Stata, SAS, Minitab, Python; descriptive and inferential statistics; ANOVA; Qualtrics; mixed methods; field data collection

Research and tools: IRB/CITI human subjects research; Technology Acceptance Model; Diffusion of Innovations; LangGraph, Streamlit, retrieval-augmented generation

Languages: Kinyarwanda (native), English (fluent), French (intermediate)

SELECTED FELLOWSHIPS, HONORS, AND SERVICE

Google Data Center Community AI Fellow, Watson Institute (2026); Emerging Leaders in AI, Black in AI (2026–2027); Bayer University Mentoring Program (2026); Global Greengrants Fund Award, $5,000 (2024); Rural Fellow, Rural Prosperity Nebraska (2021); Millennium Fellow, United Nations Academic Impact (2020)

Best African Young Achiever, Agribusiness, Young Achievers Africa Awards (2024/2025); Youth AgriChampion Award, Ban Ki-moon Centre for Global Citizens (2022)

Reviewer, ASABE journals (2026–present); Global Youth Representative, World Food Forum (2023–present); Member, UN Committee on World Food Security Data Policy group (2023–present); Steering Committee, EKYAN, UNICEF Generation Unlimited (2024–present)

References available on request.